



UCL INSTITUTE OF CHILD HEALTH
SECTION ON COGNITIVE
NEUROSCIENCE & NEUROPSYCHIATRY

The Pair Test

Administration and scoring manual



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Background

The contribution of neuropsychological evaluation to the diagnosis of memory impairment in patients with temporal lobe pathology is to 1) characterise the status of memory impairment in relation to other cognition abilities and 2) relate the selective deficits in memory to the neural systems subserving the mnemonic functions. However, available standardised tests do not allow comparable investigation of separate memory processes, for example between recall and recognition, hampering on the specificity and precision of the diagnosis. More balanced and controlled paradigms are required to investigate memory impairments in patients with pathology in the temporal lobes.

The Pair Test offers more controlled comparisons between measures, allowing to disentangle impairments in different memory processes in patients with known or suspected pathology in the temporal lobe. The advantages of the Pair Test are 1) a balanced paradigm allowing comparison between learning and memory measures, 2) assessment of learning, delayed recall and recognition using one tool, 3) paired-associate learning tool to assess cognitive processes that are dependent on the hippocampal network, and 4) the use of a tablet-based application allowing controlled administration process and engaging approach to neuropsychological testing. The test provides behavioural evidence of the integrity of the hippocampi and its interaction with the neocortical learning system. Overall, the test offers improved diagnosis, with more precise indication on the nature of the memory deficit and the underlying processing impairment.

Components of the Pair Test

The “Pair Test” is composed of analogous tasks, with relevant variables being controlled for (*i.e.* modality of presentation, item material, and access to semantic representations). Moreover, the tasks are equivalent in complexity and, as such, can be directly compared with each other to establish the specificity of memory impairment and provide a clearer understanding of memory deficits in brain-damaged patients. The Pair Test is a tablet-based Application (Figure 1), with the advantage of (1) providing a controlled administration process, (2) facilitating the administration in different contexts, and (3) increasing motivation and engagement of participants as a result of the interactive platform.

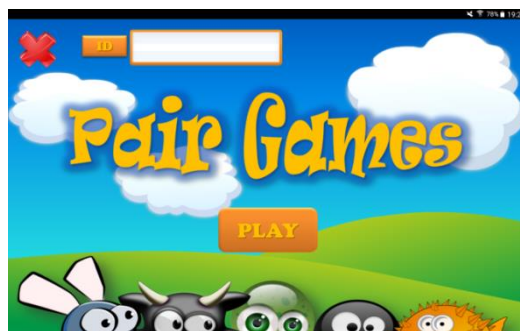
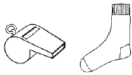


Figure 1. Interface of the Pair Test (version A)

The five analogous tasks assess verbal and nonverbal memory and learning, while controlling for modality of presentation (auditory *versus* visual) and access to semantic representations (semantic *versus* non-semantic). The five tasks are paired associates of written words, auditory words, written pseudowords, objects, and abstract designs (see

Table 1 for examples). Each task is composed of 10 pairs of stimuli to learn over 3 consecutive trials through encoding-cued recall cycles. A final recall trial and a forced-choice recognition task are administered after a 15-minute delay. The whole battery of tests requires approximately 1 ½ hours to administer. There are two versions of the Pair Test available, allowing within-subject retest at different time points.

Table 1. Example of Pairs for each subtest of the Pair Test.

Task	Example of pair
Written words	dress ocean
Auditory words	bag pin
Written pseudowords	bame ree
Objects	
Abstract designs	ooo 

These tasks assess different cognitive processes: learning, recall and recognition.

Encoding: Each pair is presented individually for 5 seconds during which the participants are asked to click on the stimulus they prefer (Figure 2). This preference judgement allows deep encoding of the pairs, minimising inter-individual variability in executive functions and controlling for concentration difficulties. This is to ensure the encoding of each pair within each task, and to use the same procedure across the 5 tasks.

Cued recall: After the last pair of the list is presented, the participant is presented with the left-side stimulus of each pair and asked to draw/write the stimulus that was paired with it (Figure 3). This encoding-cued recall cycle is repeated 3 times in a row for each task to establish a learning curve for each participant on that task.

Recognition: In the forced-choice recognition stage, participants are presented with the first stimulus of each pair (*i.e.* the cue) and asked to pick from 3 choices the target associate that was paired with the cue (Figure 4).



Figure 2. Encoding



Figure 3. Cued recall

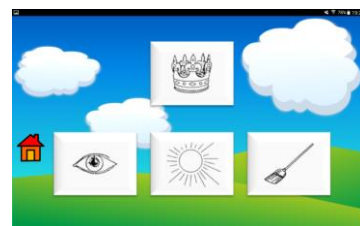


Figure 4. Recognition

Setting up

You will need an android tablet. This application will not work on Apple devices.

Download the Pair Test application from GooglePlay.

Make sure the volume is on (you should hear a music), as it will be needed for one of the tasks (*i.e.* The Spoken Words subtest).

Use the documents “Pair Test_Record form” to score performance.

Getting around the Application

There are two versions of the application:



Version A



Version B

- These are the five tasks, in each version:

-  = auditory words
-  = visual words
-  = abstract designs
-  = objects
-  = pseudowords



- To come back to this page (where you can select the task), click the  icon
-  Example when giving the instructions

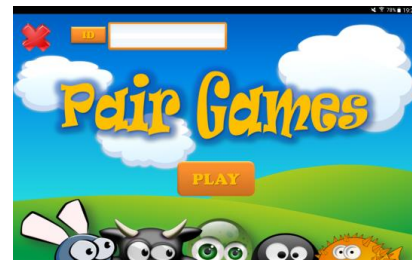
-  Encoding and 3 learning trials
 - **1** Trial 1
 - **2** Trial 2
 - **3** Trial 3
-  Delayed recall
-  Recognition



Administration

Interface

- Enter ID (the images will be saved on the device with the ID as prefix).
- Click Play.



General instructions

"You will complete five memory tests. They are very similar in that, for each, you will try and remember pairs of items; either words, objects, shapes, or made-up words. Some might be more difficult than others, but that is okay, just try your best!"

Start task

- Instructions: *"You will see/hear 10 pairs of words/objects/shapes, like this example [show example], one at a time. For each pair, click on the one you prefer. There is no right or wrong answer. After a few seconds, it will move automatically to the next pair. After we have gone through the 10 pairs, you will be shown the first word/image of each pair and you will try to remember what went with it by writing it/drawing it/saying it out loud. If you don't know, that is okay, just skip to the next pair. We will do this 3 times with the same items, so if it is a little difficult at first you may find it easier as we repeat the task. Then, after about 15minutes, we will come back to the task and you will try again to remember as many pairs as possible. Finally, we will finish we a multiple-choice where you will be given three options, and you will select the item that you think was matched to the presented item. We will do this with all 5 tasks".*
- Do not give feedback on performance.



Learning trials

- Do **1**, **2** and **3** one after the other, without breaks.

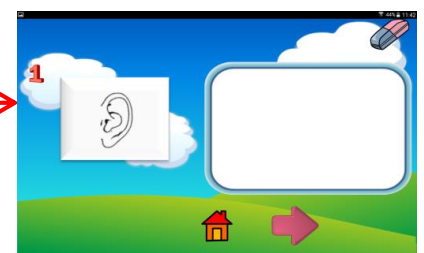
Encoding

- 10 pairs (1 shown every 5 sec)
- *"Click on the one you prefer"*



Recall

- 🌞 → say out loud (see scoring sheet!)
- 🐧 → write on canvas
- 🦇 → write on canvas
- 🐛 → draw on canvas
- 🦎 → draw on canvas



- After trial **1**: *"We will do the same thing again, with the same pairs but in a different order. Again, for each pair click on the one you prefer".*
- After trial **3**: *"I want you to keep these pairs in memory because I will ask you to remember as many as you can in about 15 minutes. In the meantime, we will start another game."*



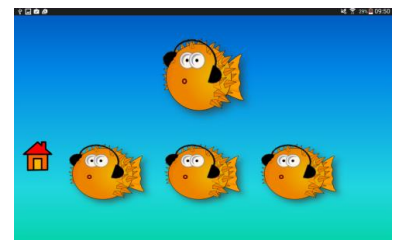
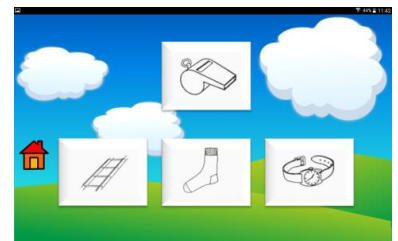
Delayed recall

- Instructions: *"Now we are going to go back to the [words/pseudowords/shapes/objects] from before. Can you try and remember as many pairs as you can?"*
- Delayed recall 15 minutes after the end of the learning trials () for that same task.
- If 15 minutes have not gone by since the end of the learning trials, then have a small break.



Recognition

- Instructions: *"The item on the top was the one that was presented on the left earlier. I want you to try and find the item that was paired with the one on top, by selecting your answer out of the three options below".*
- Do the recognition task right after the delayed recall task
- 3 choices, click on your response
- Recognition for the auditory words is a little bit different
 - Instructions: *"Click on each fish to hear the word they each have to say. Listen to the 3 possibilities before you make your choice. Click and hold on the fish to select your answer, you will then hear a *bip* which means that you have moved on to the next pair".*



Order of administration

D

To avoid interference, the tasks should be administered by alternating between verbal and non-verbal materials. Below is an example of order of administration of the full task. Note, the encoding of the second task can done during the delay period of the first task.

	Test	Process	Tick when done
1		 1	
2		 2	
3		 3	
4		 1	
5		 2	
6		 3	
7			
8			
9		 1	
10		 2	
11		 3	
12			
13			
14		 1	
15		 2	
16		 3	
17			
18			
19		 1	
20		 2	
21		 3	
22			
23			
22			
23			

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Use the table “time” to write down the time at which the learning phase is finished, for each task. The delayed recall and recognition should be done 15min after that time.

Scoring

Please find below a step-by-step description on how to score the performance on the Pair Test. An example of a completed record form is provided below, with the corresponding letters to guide through the steps.

1. Score each answer referring to the answer sheet (Version A or B).

- Correct
- Incorrect – 5 types of errors:
 - DK**: no answer is provided
 - Semantic**: the answer is semantically related to the target answer e.g. “house” for “home”
 - Form**: the answer is looks/sounds like the target answer e.g. “mouse” for “mouth” or))) for (((
 - From list**: the answer is another item from the list, but not the target answer (refer to the document “Full list of pairs”)
 - Other**: a new item, invented by the participant



Pseud	Trial 1	Trial 2	Trial 3	Delayed
Pair 1	✓	list	✓	list
Pair 2	form	✓	✓	✓
Pair 3	list	✓	✓	✓
Pair 4	new	new	✓	✓
Pair 5	✓	✓	✓	✓
Pair 6	DK	form	✓	list
Pair 7	new	list	✓	list
Pair 8	new	✓	✓	✓
Pair 9	new	DK	✓	✓
Pair 10	new	✓	list	new
	2/10	5/10	9/10	6/10

2. Calculate the average learning score.

This is simply the average across all three learning trials, for each task separately. There is a dedicated box for this score, underneath each scoring table.



Written Words				
	Trial 1	Trial 2	Trial 3	Delayed
Pair 1	V	DK	V	new
Pair 2	DK	V	V	V
Pair 3	DK	V	V	V
Pair 4	DK	DK	DK	V
Pair 5	V	V	V	DK
Pair 6	V	semantic	V	V
Pair 7	DK	V	DK	V
Pair 8	DK	V	V	V
Pair 9	V	V	V	DK
Pair 10	DK	DK	DK	V
	4/10	6/10	7/10	7/10
Average learning		5.7		

3. For each task, transfer the number of errors, in the “types of errors” table found in the record form

F

	DK	Semantic	Form	From list	Other
SW	1				
WW	3				1
Pseudo	2		2	7	7
Obj					
Des	2			5	21

A 4. Convert raw scores to composite scores (index scores)

Write the raw scores, for each subtest, in the corresponding tables. Do this for learning, recall and recognition:

- Learning: average of the three learning trials
- Delayed recall: score at delayed recall
- Delayed recognition: correct response minus familiar and new errors (C-F-N) G

Then, times those numbers by 10 and write them in the composite score boxes.

Finally, calculate the composite scores by taking the average of the corresponding raw scores.

B 5. Convert composite scores to standard scores

Refer to the standardisation tables to convert raw scores into standard scores. Do this for learning, recall and recognition.

Do this also for the general scores (general learning/recall, general recognition, general memory scores).

- General learning/recall: average between general learning composite score and general recall composite score (across all tasks).
- General recognition: general recognition composite score (box C)
- General memory: average between general learning composite score (box A), general recall composite score (box B), and general recognition composite score (box C).

C



The Pair Test

A computerised measure of learning and memory

Record Form

Examinee name: *John Smith*

Date: *24.03.2020*

Examinee ID: *873*

Examiner name: *Sarah Buck*

Age: *14*

Application version: ☒ A ☐ B

Sex: ☒ M ☐ F

Learning raw scores to composite scores (index scores)

Subtest	Raw score ⁺	Composite scores				
Spoken words	7					70
Written words	8	80		80		80
Pseudowords	5	50			50	50
Objects	8		80	80		80
Designs	6		60		60	6
Composite scores*		65	7	8	5.5	A 6.8
+average of 3 learning trials		Verbal comp.	Visual comp.	Semantic comp.	Non-semantic comp.	General learning comp.
*average of raw scores						

Composite scores to standard scores

Index	Comp. score	Std. score	
Verbal comp.	65	109	Verbal learning
Visual comp.	70	115	Visual learning
Semantic comp.	80	111	Semantic learning
Non-semantic comp.	55	118	Non-semantic learning

Delayed recall raw scores to composite scores (index scores)

Subtest	Raw score	Composite scores				
Spoken words	6					60
Written words	7	70		70		70
Pseudowords	6	60			60	60
Objects	10		100	100		100
Designs	5		50		50	50
Composite scores*		65	75	85	55	B 67
*average of raw scores		Verbal comp.	Visual comp.	Semantic comp.	Non-semantic comp.	General recall comp.

Composite scores to standard scores

Index	Comp. score	Std. score	
Verbal comp.	65	102	Verbal recall
Visual comp.	75	109	Visual recall
Semantic comp.	85	105	Semantic recall
Non-semantic comp.	55	107	Non-semantic recall

Delayed recognition raw scores to composite scores (index scores)

Subtest	Raw score	Composite scores				
Spoken words	10					100
Written words	8	80		80		80
Pseudowords	6	60			60	60
Objects	10		100	100		100
Designs	6		60		60	60
Composite scores*		70	80	90	60	C 80
*average of raw scores		Verbal comp.	Visual comp.	Semantic comp.	Non-semantic comp.	General recogn. comp.

Composite scores to standard scores

Index	Comp. score	Std. score	
Verbal comp.	70	88	Verbal recognition
Visual comp.	80	98	Visual recognition
Semantic comp.	90	100	Semantic recognition
Non-semantic comp.	60	89	Non-semantic recognition

C

Composite scores to standard scores

Index	Comp. score	Std. score
General learning/recall	67	107
General recognition	80	97
General memory	72	105

To obtain this comp. score, calculate the average between A (general learning comp.) and B (general recall comp.)

To obtain this comp. score, calculate the average between A (gen. learning comp.), B (gen. recall comp.), and C (gen. recogn. comp.)

D

Order of administration of tasks

Subtest	Order
Spoken words	3
Written words	1
Pseudowords	5
Objects	2
Designs	4

E

Time

+15 min



	Task 1	Task 2	Task 3	Task 4	Task 5
End	1.12	1.25	1.38	2.03	2.16
Start	1.27	1.40	1.53	2.18	2.31

F

Types of errors

Subtest	DK	Semantic	Form	From list	Other
Spoken words	13	2	0	4	1
Written words	14	1	0	0	1
Pseudowords	10	0	4	2	5
Objects	4	0	0	1	1
Designs	10	0	5	4	3

Recognition scores

Subtest	Correct	Familiar error	New error	Score (C-F-N)	Time (sec)
Spoken words	10	0	0	10	43
Written words	9	1	0	8	28
Pseudowords	8	1	1	6	31
Objects	10	0	0	10	26
Designs	8	1	1	6	33

G

Spoken Words

H	 A	Trial 1			Trial 2			Trial 3			Delayed recall		
	Pair 1	fox	rabbit	V	bag	pin	barrel	face	Jacket	V	bag	pin	V
	Pair 2	pool	Wood	DK	uncle	square	DK	pool	Wood	V	uncle	square	V
	Pair 3	teacher	bubble	V	knife	fork	DK	girl	Barrel	DK	knife	fork	DK
	Pair 4	face	jacket	coat	face	jacket	V	bag	Pin	V	face	jacket	V
	Pair 5	knife	Fork	DK	pool	wood	V	knife	Fork	spoon	pool	wood	V
	Pair 6	girl	barrel	V	garbage	Market	boy	turtle	Smoke	V	garbage	Market	pin
	Pair 7	uncle	square	pin	teacher	bubble	V	uncle	Square	V	teacher	bubble	V
	Pair 8	turtle	smoke	DK	fox	rabbit	V	fox	Rabbit	V	fox	rabbit	V
	Pair 9	bag	pin	DK	girl	barrel	V	garbage	Market	DK	girl	Barrel	DK
	Pair 10	garbage	market	DK	turtle	Smoke	DK	teacher	Bubble	fork	turtle	Smoke	DK

3/10

5/10

6/10

6/10

I

Average learning

7

Written Words					Pseudowords				
	Trial 1	Trial 2	Trial 3	Delayed		Trial 1	Trial 2	Trial 3	Delayed
Pair 1	V	DK	V	new	Pair 1	V	DK	V	form
Pair 2	DK	V	V	V	Pair 2	V	V	list	DK
Pair 3	DK	V	V	V	Pair 3	new	new	V	V
Pair 4	DK	DK	DK	V	Pair 4	form	form	DK	V
Pair 5	V	V	V	DK	Pair 5	V	V	V	new
Pair 6	V	semantic	V	V	Pair 6	DK	DK	V	V
Pair 7	DK	V	DK	V	Pair 7	list	V	new	V
Pair 8	DK	V	V	V	Pair 8	DK	DK	V	V
Pair 9	V	V	V	DK	Pair 9	DK	V	V	new
Pair 10	DK	DK	DK	V	Pair 10	DK	DK	form	V
	4/10	6/10	7/10	7/10		3/10	4/10	6/10	6/10
Average learning		5.7			Average learning		4.3		

Objects					Designs				
	Trial 1	Trial 2	Trial 3	Delayed		Trial 1	Trial 2	Trial 3	Delayed
Pair 1	V	V	V	V	Pair 1	V	form	V	list
Pair 2	new	V	V	V	Pair 2	V	V	new	DK
Pair 3	V	V	V	V	Pair 3	DK	list	V	V
Pair 4	list	DK	V	V	Pair 4	DK	new	form	V
Pair 5	V	V	V	V	Pair 5	V	V	list	DK
Pair 6	V	DK	V	V	Pair 6	DK	DK	V	V
Pair 7	V	V	V	V	Pair 7	V	V	DK	V
Pair 8	DK	V	V	V	Pair 8	DK	form	V	V
Pair 9	V	V	V	V	Pair 9	list	V	V	form
Pair 10	DK	V	V	V	Pair 10	DK	form	DK	new
	6/10	8/10	10/10	10/10		4/10	4/10	5/10	5/10
Average learning		8			Average learning		4.3		